

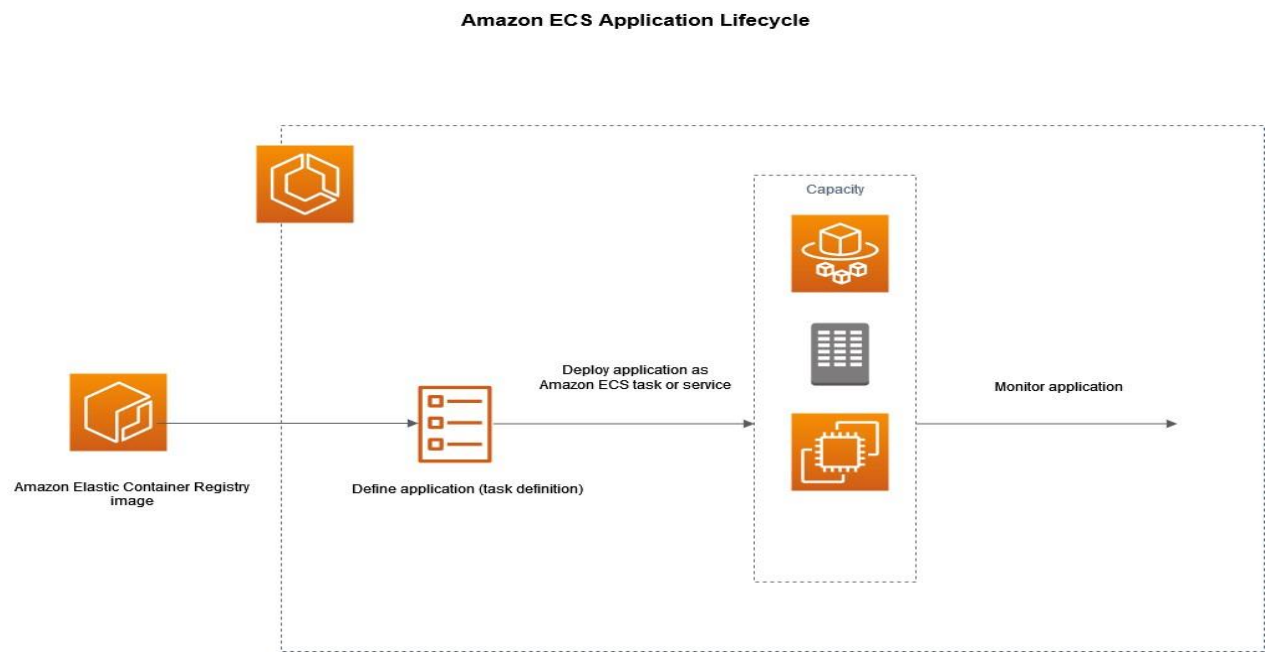
Deploy Serverless Applications

ECS-[Elastic Container Service]

Amazon Elastic Container Service (Amazon ECS) : is a fully managed container orchestration service that helps you easily deploy, manage, and scale containerized applications.

Application lifecycle :

The following diagram shows the application lifecycle and how it works with the Amazon ECS components.



Steps

Launch linux instance

EC2 > Instances > Launch an instance

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

myLinux

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Summary

Number of instances

Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.6.2...read more

ami-0fd05997b4dff7aac

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

Cancel

Launch instance

Preview code

Connect with instance

EC2 > Instances > i-063fcc5e4bb8c12a4 > Connect to instance

Connect to instance

Info

Connect to your instance i-063fcc5e4bb8c12a4 (myLinux01) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 serial console

Instance ID

i-063fcc5e4bb8c12a4 (myLinux01)

1. Open an SSH client.

2. Locate your private key file. The key used to launch this instance is LinuxKeyPair.pem

3. Run this command, if necessary, to ensure your key is not publicly viewable.

chmod 400 "LinuxKeyPair.pem"

4. Connect to your instance using its Public DNS:

ec2-3-108-51-27.ap-south-1.compute.amazonaws.com

Example:

ssh -i "LinuxKeyPair.pem" ec2-user@ec2-3-108-51-27.ap-south-1.compute.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

- --> Install Docker


```
root@ip-172-31-3-247:/home/ec2-user
[root@ip-172-31-3-247 ec2-user]# service docker start
Redirecting to /bin/systemctl start docker.service
[root@ip-172-31-3-247 ec2-user]# vi Dockerfile
[root@ip-172-31-3-247 ec2-user]# vi index.html
[root@ip-172-31-3-247 ec2-user]# cat index.html
<html>
<h1>Welcome to ECR</h1>
</html>

[root@ip-172-31-3-247 ec2-user]# cat Dockerfile
FROM ubuntu
RUN apt update -y
RUN apt-get install apache2 -y
COPY index.html /var/www/html
EXPOSE 80
CMD ["apache2ctl", "-D", "FOREGROUND"]

[root@ip-172-31-3-247 ec2-user]# docker build -t img1:v1 .
[+] Building 26.8s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> -- transferring Dockerfile: 216B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> -- transferring context: 2B
=> [1/4] FROM docker.io/library/ubuntu:latest@sha256:80dd1c3b9c6c0cb8f1667e8290b3bc61b78c2678c02cbdae5f0fea92cc6734ab
=> -- resolve docker.io/library/ubuntu:latest@sha256:80dd1c3b9c6c0cb8f1667e8290b3bc61b78c2678c02cbdae5f0fea92cc6734ab
=> -- sha256:80dd1c3b9c6c0cb8f1667e8290b3bc61b78c2678c02cbdae5f0fea92cc6734ab 6.49kB / 6.49kB
=> -- sha256:6e75a106f2c0b5a2763e8118a369cc7cc304fa1b6748c491bb21f18c3e7 424B / 424B
=> -- sha256:c1d1dd784b8155984794e522b380878cf9a32d4c1fb67293cfc931c3aa6ae4 2.10kB / 2.10kB
=> -- sha256:de44b26597ae44b212de1cb896848666f13b35c1090d9709060bc8e1bb2d64 29.75MB / 29.75MB
=> -- extracting sha256:de44b26597ae44b212de1cb896848666f13b35c1090d9709060bc8e1bb2d64
=> [internal] load build context
=> -- transferring context: 139B
=> [2/4] RUN apt update -y
=> [3/4] RUN apt-get install apache2 -y
=> [4/4] COPY index.html /var/www/html
=> exporting to image
=> -- exporting layers
=> -- writing image sha256:715975a90057cd8b1040f288ad37a3f06dbcf946fd260d35e3f77e47aa2a70c
=> -- naming to docker.io/library/img1:v1
[root@ip-172-31-3-247 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
img1 v1 715975a90057 13 seconds ago 229MB
[root@ip-172-31-3-247 ec2-user]#
```

--> create repository in ECR--> provide name

Amazon ECR > Private registry > Repositories

Amazon Elastic Container Registry

Private registry

Repositories

Features & Settings

Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

Getting started

Successfully created myrepo11

Private repositories (1)

View push commands Delete Actions Create repository

Search by repository substring

Repository name	URI	Created at	Tag immutability	Encryption type
myrepo11	637423422597.dkr.ecr.ap-south-1.amazonaws.com/myrepo11	December 29, 2024, 17:01:44 (UTC+05.5)	Mutable	AES-256

--> goin repo name --> select viewpush command nd copy paste the commands inside putty after configuring AWS

Amazon Elastic Container Registry

▼ **Private registry**

Repositories

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▼ **Public registry**

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ECR public gallery

 Image scan overview, status, and full vulnerabilities has moved to the Image detail page. To access, click an image tag.

Images (0)

 Search artifacts

 Image tag

Artifact type

Pushed at

Size (MB)

Image URI

Digest

No images

No images to display

Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see [Registry Authentication](#).

1. Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:

```
aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin  
637423422597.dkr.ecr.ap-south-1.amazonaws.com
```

Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.

2. Build your Docker image using the following command. For information on building a Docker file from scratch see the instructions [here](#). You can skip this step if your image is already built:

```
❏ docker build -t myrepo11 .
```

3. After the build completes, tag your image so you can push the image to this repository:

```
❏ docker tag myrepo11:latest 637423422597.dkr.ecr.ap-south-1.amazonaws.com/myrepo11:latest
```

4. Run the following command to push this image to your newly created AWS repository:

```
docker push 637423422597.dkr.ecr.ap-south-1.amazonaws.com/myrepo11:latest
```

 Copied 

Close

```
root@ip-172-31-3-247:/home/ec2-user
-> [internal] load metadata for docker.io/library/ubuntu:latest
-> [internal] load .dockerignore
-> transferring context: 2B
-> [1/4] FROM docker.io/library/ubuntu:latest@sha256:90dd3c3b9e6e9eb9f1667e9290b3bc61b79c2678c02cbdae5f0fea92cc6734ab
-> resolve docker.io/library/ubuntu:latest@sha256:90dd3c3b9e6e9eb9f1667e9290b3bc61b79c2678c02cbdae5f0fea92cc6734ab
-> sha256:90dd3c3b9e6e9eb9f1667e9290b3bc61b79c2678c02cbdae5f0fea92cc6734ab 6.69kB / 6.69kB
-> sha256:ce75a10070b0fc0bead763c5118a36be7c10dfc1b0749c491bb21f15c3c7 424B / 424B
-> sha256:b1d9df8ab1559494794e532b380978cf9ba82d4c1fb67293bcf931c1aa69ae4 2.30kB / 2.30kB
-> sha256:de44b265597ae44b212defcb50694d666f13b35c1090d9709069bc861bb2d64 29.75MB / 29.75MB
-> extracting sha256:de44b265597ae44b212defcb50694d666f13b35c1090d9709069bc861bb2d64
-> [internal] load build context
-> transferring context: 139B
-> [2/4] RUN apt update -y
-> [3/4] RUN apt-get install apache2 -y
-> [4/4] COPY index.html /var/www/html
-> exporting to image
-> exporting layers
-> writing image sha256:715975a90057c8b1040f288ad37a3f06dbcf946fd260dd5e3f77e47aa2a70c
-> naming to docker.io/library/img1:v1
[root@ip-172-31-3-247 ec2-user]# docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
img1 v1 715975a90057 13 seconds ago 229MB
[root@ip-172-31-3-247 ec2-user]# aws configure
AWS Access Key ID [None]: AKIAJ2I2LFRSCYR0XJVKV
AWS Secret Access Key [None]: gsiV5qd4sybWpaz6VfU43pOBWRKtYGFvTbBh/ECW
Default region name [None]:
Default output format [None]:
[root@ip-172-31-3-247 ec2-user]#
[root@ip-172-31-3-247 ec2-user]# aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 637423422597.dkr.ecr.ap-south-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-3-247 ec2-user]# docker tag img1:v1 637423422597.dkr.ecr.ap-south-1.amazonaws.com/myrepo11:v1
[root@ip-172-31-3-247 ec2-user]# docker push 637423422597.dkr.ecr.ap-south-1.amazonaws.com/myrepo11:v1
The push refers to repository [637423422597.dkr.ecr.ap-south-1.amazonaws.com/myrepo11]
681e1057e708: Pushed
aacc05a27c1c: Pushed
8f391455f10a: Pushed
687d50f2f6a6: Pushed
v1: digest: sha256:0ba05f82bc0ba8a5c30c541474bbce2ea588b66c5afalacbb3e407636781c9c1 size: 1160
[root@ip-172-31-3-247 ec2-user]#
```

Amazon ECR > Private registry > Repositories > myrepo11

Amazon Elastic Container Registry

Private registry

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ECR public gallery

Image scan overview, status, and full vulnerabilities has moved to the Image detail page. To access, click an image tag.

Images (1)

Search artifacts

Image tag

Artifact type

Pushed at

Size (MB)

Image URI

Digest

☐

v1

Image

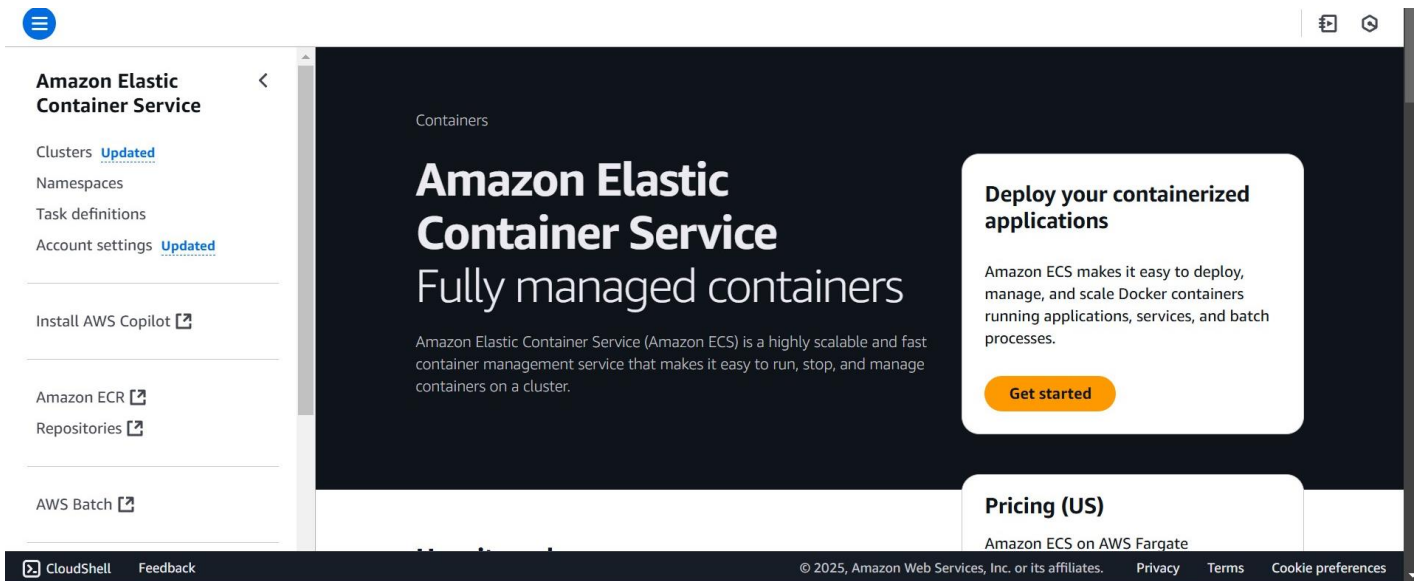
December 29, 2024, 17:15:31 (UTC+05.5)

97.29

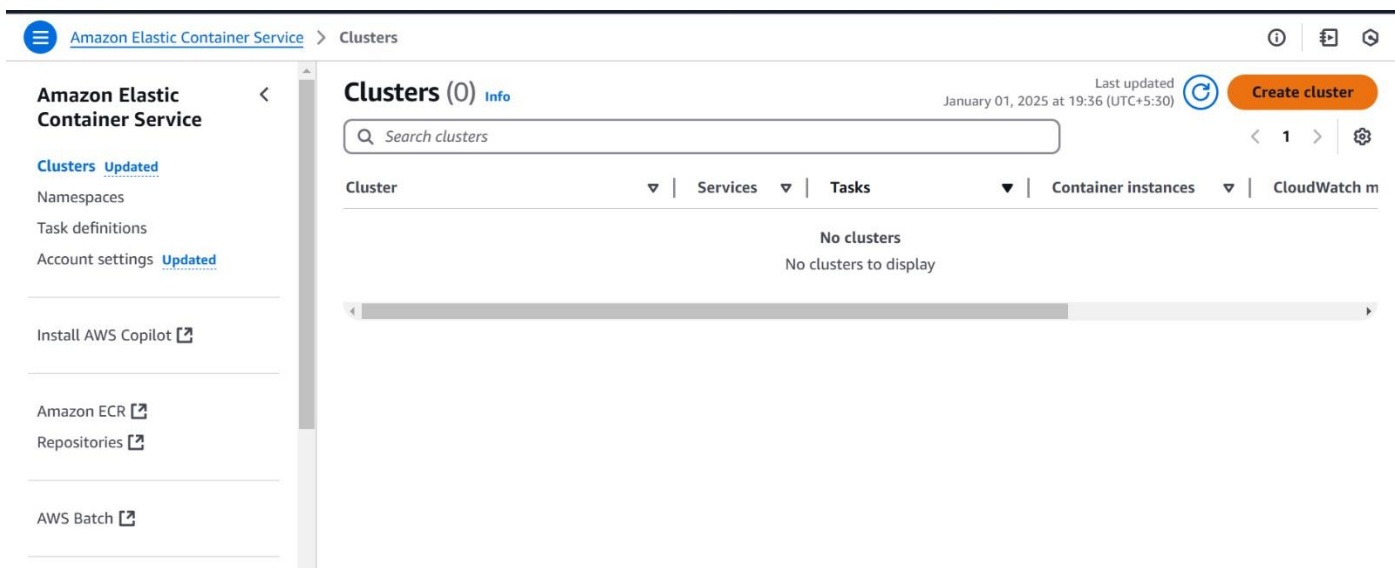
Copy URI

sha256:0ba05f8...

close putty --> -->
Search ECS



--> Create Cluster



-->provide name

-->under Infrastructure select AWS fargate which is serverless

Amazon Elastic Container Service > Create cluster

and capacity must belong to a cluster.

Cluster configuration

Cluster name

 Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Default namespace - optional
 Select the namespace to specify a group of services that make up your application. You can overwrite this value at the service level.

▼ **Infrastructure** [Info](#) Serverless

Your cluster is automatically configured for AWS Fargate (serverless) with two capacity providers. Add Amazon EC2 instances.

☒ **AWS Fargate (serverless)**
 Pay as you go. Use if you have tiny, batch, or burst workloads or for zero maintenance overhead. The cluster has Fargate and Fargate Spot capacity providers by default.

☐ **Amazon EC2 instances**

--> Select Create--> Go in Task defination--> crate new task defn 2--> give family name--> choose launch type as AWS farget

Amazon Elastic Container Service > Create new task definition

Cluster mycluster13 has been created successfully. [View cluster](#)

Create new task definition

Task definition configuration

Task definition family [Info](#)
 Specify a unique task definition family name.

 Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

▼ **Infrastructure requirements**
 Specify the infrastructure requirements for the task definition.

Launch type [Info](#)
 Selection of the launch type will change task definition parameters.

☒ **AWS Fargate**
 Serverless compute for containers.

--> Inside Task size select 0.5CPU and 1 gb memory-->

Amazon Elastic Container Service > Create new task definition

OS, Architecture, Network mode
Network mode is used for tasks and is dependent on the compute type selected.

Operating system/Architecture | Info **Network mode** | Info

Linux/X86_64 awsvpc

Task size | Info
Specify the amount of CPU and memory to reserve for your task.

CPU **Memory**

.5 vCPU 1 GB

▼ **Task roles - conditional**

Task role | Info
A task IAM role allows containers in the task to make API requests to AWS services. You can create a task IAM role from the [IAM console](#).

-

Task execution role | Info
A task execution IAM role is used by the container agent to make AWS API requests on your behalf. If you don't already have a task execution IAM role created, we can create one for you.

ecsTaskExecutionRole

--> Inside Container-1 provide Container name

Amazon Elastic Container Service > Create new task definition

Amazon Elastic Container Service

Clusters Updated
Namespaces
Task definitions
Account settings Updated

Install AWS Copilot

Amazon ECR
Repositories

AWS Batch

▼ **Container - 1** | Info Essential container Remove

Container details
Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name **Image URI** **Essential container**

Cont-1 public.ecr.aws/nginx/nginx:stable-perl Yes

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed. Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

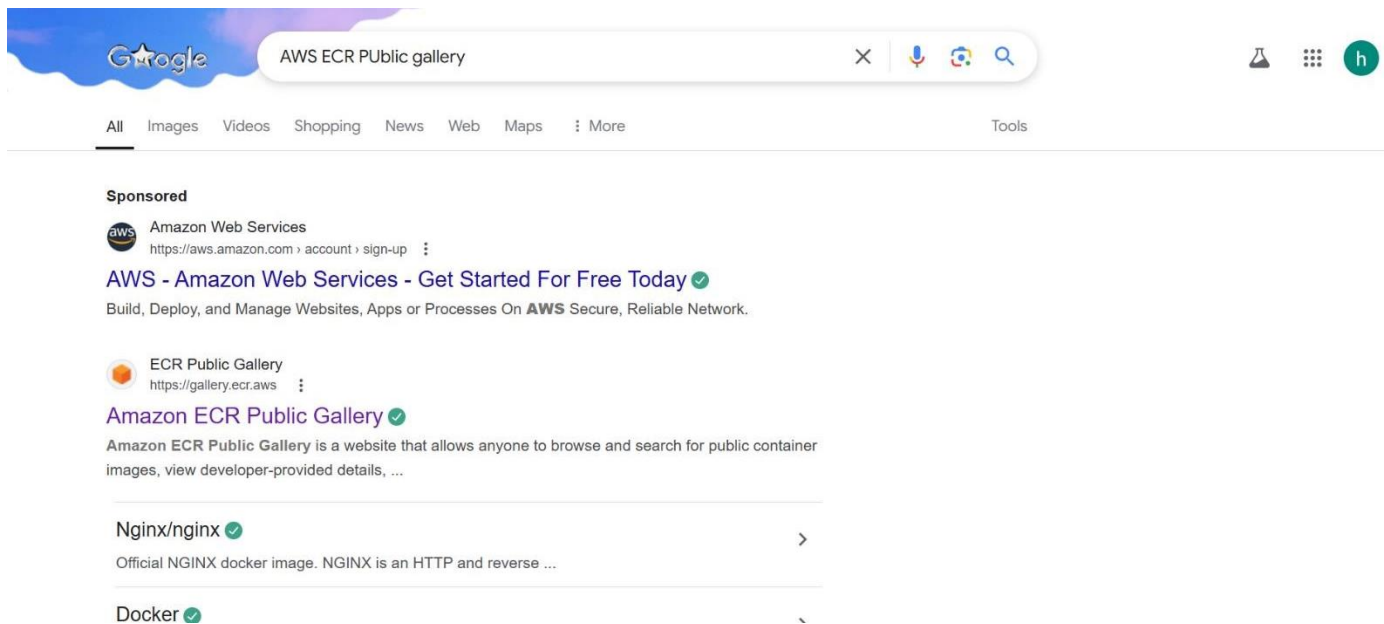
Private registry | Info
Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.

☐ Private registry authentication

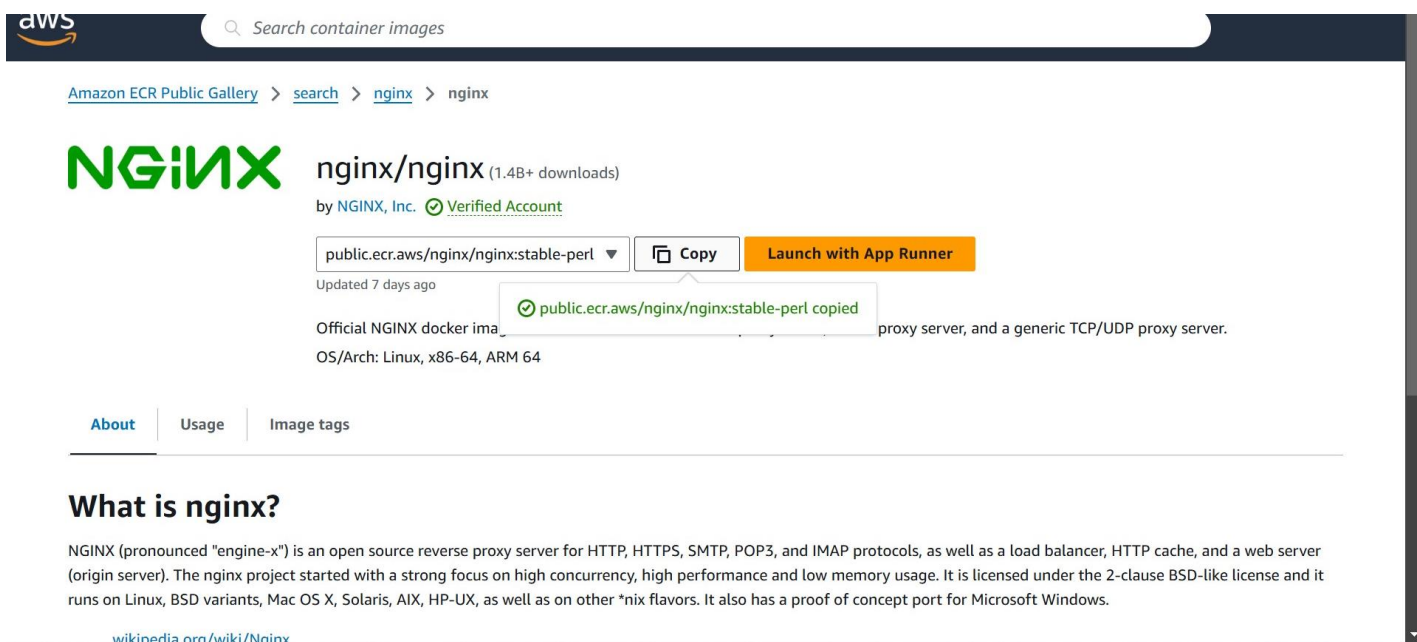
Port mappings | Info
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Container port	Protocol	Port name	App protocol	
80	TCP	container-port-prot	HTTP	Remove

--> Search AWS ECR public Gallery



--> Select NGiNX--> select the first link--> COpy the URI--> and paste it in Image URI



--> Under the logging untick use log collection

Amazon Elastic Container Service

Create new task definition

Amazon Elastic Container Service

Clusters Updated

Namespaces

Task definitions

Account settings Updated

Install AWS Copilot

Amazon ECR

Repositories

AWS Batch

Add environment file

You can add 10 more environment files.

Logging - optional

CPU and memory allocation for a sidecar

There are logging options that will automatically add a sidecar to your task definition if it does not already exist. AWS provides CPU and memory adjustment recommendations based on the selected options.

We recommend that you use log collection for tasks running on AWS Fargate. Learn more about log collection.

Log collection

Configure your task to send container logs to a logging destination using a default configuration. See pricing information on Amazon CloudWatch.

Use log collection

Restart policy - optional

Configure a container restart policy to restart individual containers locally without restarting the whole task.

HealthCheck - optional

--> Create.

Amazon Elastic Container Service

Create new task definition

Amazon Elastic Container Service

Clusters Updated

Namespaces

Task definitions

Account settings Updated

Install AWS Copilot

Amazon ECR

Repositories

AWS Batch

Add volume

Volumes from

Mount data volumes from another container.

Add volume from

Monitoring - optional

Configure your application trace and metric collection settings using the AWS Distro for OpenTelemetry integration.

Tags - optional

Tags help you to identify and organize your task definitions.

Cancel

Create

--> Go inside the Cluster name

Amazon Elastic Container Service > Clusters > mycluster13 > Services

Task definition successfully created
TD13:1 has been successfully created. You can use this task definition to deploy a service or run a task. [View task definition](#)

mycluster13 Last updated January 01, 2025 at 19:55 (UTC+5:30) [Update cluster](#) [Delete cluster](#)

Cluster overview

ARN arn:aws:ecs:ap-south-1:637423422597:cluster/mycluster13	Status Active	CloudWatch monitoring Default	Registered container instances -
Services Draining - Active - Encryption		Tasks Pending - Running -	

-> scroll below and select Create service

Amazon Elastic Container Service > Clusters > mycluster13 > Services

Fargate ephemeral storage
-

Services | Tasks | Infrastructure | Metrics | Scheduled tasks | Tags

Services (0) [Info](#) [Manage tags](#) [Update](#) [Delete service](#) [Create](#)

Filter launch type: Any launch type | Filter service type: Any service type

Filter services by value:

Service name | ARN | Status | Service... | Deployments and tasks

No services
No services to display.
[Create](#)

--> under deployment configuration select our task family

Amazon Elastic Container Service > Clusters > mycluster13 > Create service

Deployment configuration

Application type [Info](#)
Specify what type of application you want to run.

☐ Service
Launch a group of tasks handling a long-running computing work that can be stopped and restarted. For example, a web application.

☒ Task
Launch a standalone task that runs and terminates. For example, a batch job.

Task definition
Select an existing task definition. To create a new task definition, go to [Task definitions](#).

☐ Specify the revision manually
Manually input the revision instead of choosing from the 100 most recent revisions for the selected task definition family.

Family: TD13 [Refresh](#) **Revision**: 1 (LATEST)

Desired tasks
Specify the number of tasks to launch.
1

--> Provide Service Name--> Under Networking Select Linux Security Group

Amazon Elastic Container Service > Clusters > mycluster13 > Create service

Choose the subnets within the VPC that the task scheduler should consider for placement.

Choose subnets Clear current selection

subnet-0c8dde09bdd7c8eac X
ap-south-1c 172.31.16.0/20

subnet-0bed7e95e010d2720 X
ap-south-1b 172.31.0.0/20

subnet-018030706c398b32a X
ap-south-1a 172.31.32.0/20

Security group | Info
Choose an existing security group or create a new security group.

☒ Use an existing security group
☐ Create a new security group

Security group name
Choose an existing security group.

Choose security groups

sg-070b5824b9aedc64b X
LinuxSecurityGroup

Public IP | Info
Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).

--> Expand Load balancer--> select Application Load balancer--> Provide Load balancer Name

Amazon Elastic Container Service > Clusters > mycluster13 > Create service

▼ Load balancing - optional
Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

Load balancer type | Info
Configure a load balancer to distribute incoming traffic across the tasks running in your service.

Application Load Balancer

Container
The container and port to load balance the incoming traffic to

Cont-1 80:80

Host port:Container port

Application Load Balancer
Specify whether to create a new load balancer or choose an existing one.

☒ Create a new load balancer
☐ Use an existing load balancer

Load balancer name
Assign a unique name for the load balancer.

--> Provide Target Group name

Amazon Elastic Container Service > Clusters > mycluster13 > Create service

Manually input the revision instead of choosing from the 100 most recent revisions for the selected task definition family.

Family TD13 **Revision** 1 (LATEST)

Service name
Assign a service name that is unique for this cluster.
myService13
Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.

Service type [Info](#)
Specify the service type that the service scheduler will follow.

☒ **Replica**
Place and maintain a desired number of tasks across your cluster.

☐ **Daemon**
Place and maintain one copy of your task on each container instance.

Desired tasks
Specify the number of tasks to launch.
1

--> and create

Amazon Elastic Container Service > Clusters > mycluster13 > Create service

VPC Lattice - optional, new [Info](#)
Fully managed application networking service to connect, secure, and monitor your services across multiple accounts and virtual private clouds (VPCs). When you use VPC Lattice, there is a cost associated with it.

Service auto scaling - optional
Automatically adjust your service's desired count up and down within a specified range in response to CloudWatch alarms. You can modify your service auto scaling configuration at any time to meet the needs of your application.

Volume - optional [Info](#)
Configure a data volume to provide additional storage for the containers in the task.

Tags - optional [Info](#)
Tags help you to identify and organize your resources.

[Cancel](#) [Create](#)

--> After Service is Created go in load balancer Name Inside EC2 Service

EC2 > Load balancers

Load balancers (1) [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☐	Name	DNS name	State	VPC ID	Availability Zones
☐	LB13	LB13-1610048338.ap-sout...	Active	vpc-09867797ad23c076c	3 Availability Zones

0 load balancers selected

Select a load balancer above.

--> Copy the DNS name and paste it in New Tab

The screenshot shows the AWS Management Console interface for an Elastic Load Balancing (ELB) instance named LB13. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, and AMI Catalog. The main content area displays the following details:

- Scheme:** Internet-facing
- Hosted zone:** ZP97RAFLXTNZK
- Availability Zones:** Three subnets are listed: subnet-0c8dde09bdd7c8eac (ap-south-1c (aps1-az2)), subnet-0bed7e95e010d2720 (ap-south-1b (aps1-az3)), and subnet-018030706c398b32a (ap-south-1a (aps1-az1)).
- Date created:** January 1, 2025, 20:11 (UTC+05:30)
- Load balancer ARN:** arn:aws:elasticloadbalancing:ap-south-1:637423422597:loadbalancer/app/LB13/83abf19f94577e2a
- DNS name:** LB13-1610048338.ap-south-1.elb.amazonaws.com (A Record)

A green notification bubble indicates "DNS name copied". Below the details, there are tabs for Listeners and rules, Network mapping, Resource map - new, Security, Monitoring, and Integrations. The "Listeners and rules" tab is active, showing "Listeners and rules (1) Info" and buttons for "Manage rules", "Manage listener", and "Add listener".

--> Here u will see the Welcome Page of nginx

The screenshot shows a web browser window with the address bar displaying "lb13-1610048338.ap-south-1.elb.amazonaws.com". The page content is as follows:

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

The browser's taskbar at the bottom shows various application icons, including the Start menu, Search, and several open applications. The system tray on the right indicates the language is "ENG IN", the date is "2015 01-01-2025", and the time is "20:15".